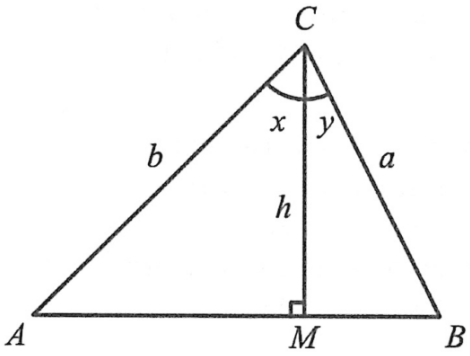
Trigonometry: Compound Angle Formulae

Questions 1–36

Question 1. The diagram below shows the triangle ABC. It is given that angle ACM is x degrees and angle BCM is y degrees. The length of BC is a and the length of AC is b.



a) Write h in terms of b and x.

b) Write h in terms of a and y.

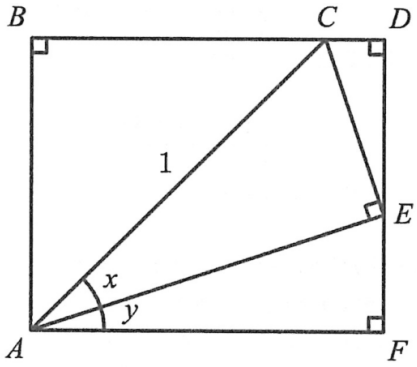
c) Find the area of triangles ACM and BCM.

d) Find the area of triangle ABC.

e) Hence show that

$$\sin(x+y) = \sin x \cos y + \cos x \sin y.$$

Question 2. The diagram below shows the rectangle $ABDF$. It is given that angle CAE is x degrees and angle EAF is y degrees. It is also given that the length AC is 1 unit.



a) Find the lengths of AE & CE in terms of x.

b) Find the angles ACE, AEF, CED, ECD, BCA and BAC in terms of x and y.

c) Find the lengths of BC and AB in terms of x and y.

d) Find the lengths of AF and EF in terms of x and y.

e) Find the lengths of DE and CD in terms of x and y.

f) Use the fact that $AB = DE + EF$ and $BC = AF - CD$ to write down two compound angle formulae.

Question 3. Given $\sin(x+y) = \sin x \cos y + \cos x \sin y$, show that $\sin(x-y) = \sin x \cos y - \cos x \sin y$.

Question 4. Given $\cos(x+y) = \cos x \cos y - \sin x \sin y$, show that $\cos(x-y) = \cos x \cos y + \sin x \sin y$.

Question 5. By writing $\tan(x+y)$ as $\frac{\sin(x+y)}{\cos(x+y)}$, show that

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}.$$

Question 6. Given $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$, show that

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}.$$

Question 7. Write the following in the form $\sin k$, where k is an acute angle:

a) $\sin 30° \cos 45° + \cos 30° \sin 45°$

b) $\sin 35° \cos 20° - \cos 35° \sin 20°$

c) $\sin 70° \cos 50° - \cos 70° \sin 50°$

d) $\sin 70° \cos 50° + \cos 70° \sin 50°$

Question 8. Write the following in the form $\cos k$, where k is an acute angle:

a) $\cos 50° \cos 30° + \sin 50° \sin 30°$

b) $\cos 68° \cos 13° + \sin 68° \sin 13°$

c) $\cos 10° \cos 54° - \sin 10° \sin 54°$

d) $\cos 300° \cos 50° - \sin 300° \sin 50°$

Question 9.

a) i) Write down the exact values of $\sin 30°, \sin 45°$ and $\sin 60°$

ii) Write down the exact values of $\cos 30°, \cos 45°$ and $\cos 60°$

b) Hence use a compound angle formula to find the exact value of:

i) $\sin 75°$

ii) $\sin 15°$

iii) $\cos 105°$

iv) $\cos 15°$

Question 10.

a) Write down the exact values of $\tan 30°$, $\tan 45°$ and $\tan 60°$.

b) Hence use a compound angle formula to find the exact value of:

i) $\tan 75°$

ii) $\tan 15°$

Question 11. Use the compound angle formulae to simplify each of these:

a) $\sin(x + 90°)$

b) $\cos(x - 90°)$

c) $\cos(x - 180°)$

d) $\sin(x - 270°)$

Question 12. Write the following in the form $\sin k$, where k is an acute angle:

a) $\cos 40° \cos 32° - \sin 40° \sin 32°$

b) $\cos 128° \cos 125° + \sin 128° \sin 125°$

Question 13. Write the following in the form $\cos k$, where k is an acute angle:

a) $\sin 96° \cos 5° - \cos 96° \sin 5°$

b) $\sin 25° \cos 42° + \cos 25° \sin 42°$

Question 14. It is given that $\cos x = \frac{2}{9}$ and $\cos y = \frac{6}{7}$ and that x and y are acute.
Find the exact value of:

a) $\cos(x - y)$

b) $\sin(x - y)$

Question 15. It is given that $\tan 85° = b$. Show that

$$\tan 130° = \frac{1+b}{1-b}.$$

Question 16. It is given that $\cos 55° = m$. Show that

$$\cos 5° = \frac{m + \sqrt{3 - 3m^2}}{2}.$$

Question 17. Find the minimum value of the following expressions and the positive value of θ, in degrees, for which the minimum first occurs:

a) $\sin\theta\cos 60° + \cos\theta\sin 60°$

b) $2\cos\theta\cos 35° + 2\sin\theta\sin 35°$

c) $4\sin\theta\cos 42° - 4\cos\theta\sin 42°$

Question 18. Prerequisite knowledge for Q18–Q19: Radians Find the minimum value of the following expressions and the positive value of θ, in radians, for which the minimum first occurs:

a) $\sin\theta\cos\frac{\pi}{4} + \cos\theta\sin\frac{\pi}{4}$

b) $3\cos\theta\cos\frac{5\pi}{4} + 3\sin\theta\sin\frac{5\pi}{4}$

c) $-2\cos\theta\sin 1.6 + 2\sin\theta\cos 1.6$

Question 19. Find the maximum value of the following expressions and the positive value of θ, in radians, for which the maximum first occurs:

a) $\sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$

b) $9 \cos 5\theta \cos 2\theta + 9 \sin 5\theta \sin 2\theta$

c) $-\frac{2}{3} \cos 8\theta \sin 2\theta + \frac{2}{3} \sin 8\theta \cos 2\theta$

Question 20. Solve the following equations:

a) $\sin\theta \cos 20° + \cos\theta \sin 20° = 0.5$ for $0° \leq \theta < 360°$

b) $\cos\theta \cos 40° + \sin\theta \sin 40° = 0.34$ for $0° \leq \theta < 360°$

Question 21. Write the following in the form $\tan k$, where k is an acute angle:

a) $\frac{\tan 200^\circ + \tan 34^\circ}{1 - \tan 200^\circ \tan 34^\circ}$

b) $\frac{\tan 18^\circ - \tan 130^\circ}{1 + \tan 18^\circ \tan 130^\circ}$

Question 22. Solve the following equations:

a) $\frac{\tan \theta + \tan 20^\circ}{1 - \tan \theta \tan 20^\circ} = 4$ for $0^\circ \leq \theta < 360^\circ$

b) $\frac{\tan 3\theta - \tan 33^\circ}{1 + \tan 3\theta \tan 33^\circ} = -6$ for $0^\circ \leq \theta < 120^\circ$

Question 23. Simplify the following expressions:

a) $\sin(x + 45°) + \cos(x + 45°)$

b) $\sin(x + 135°) - \cos(x + 135°)$

c) $\sin(x - 75°) - \cos(x - 75°)$

Question 24. Prerequisite knowledge for Q24: Radians Simplify the following expressions:

a) $\sin(x - \frac{\pi}{3}) + \cos(\frac{\pi}{3} - x)$

b) $\sin(\frac{\pi}{3} - x) - \cos(\frac{4\pi}{3} - x)$

c) $-\sin(\frac{\pi}{4} - x) + \cos(\frac{2\pi}{3} - x)$

Question 25. Solve the following equations:

a) $\sin(\theta + 30°) = \cos\theta$ for $0° < \theta < 360°$

b) $\sin(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$

c) $\cos(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$

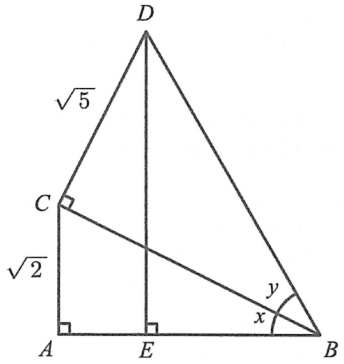
Question 26. Solve the following equations:

a) $\sin(\theta - 30°) = \cos(\theta + 45°)$ for $0° < \theta < 360°$

b) $\cos(\theta - 60°) = \cos(\theta - 45°)$ for $0° < \theta < 360°$

c) $\cos(\theta - 120°) = \sin(\theta + 60°)$ for $0° < \theta < 360°$

Question 27. In the diagram below, angle ABC is x degrees and angle CBD is y degrees. The lengths of AC and CD are $\sqrt{2}$ and $\sqrt{5}$ respectively. Show that $DE = \sqrt{2} + \sqrt{5}\cos x$.



Question 28. Prove the following trigonometric identity

$$\frac{\cos(x+y)}{\cos x \cos y} \equiv 1 - \tan x \tan y.$$

Question 29. Solve the equation

$$\frac{\tan\theta - \tan 30^\circ}{1 + \tan\theta \tan 30^\circ} = 2$$

for $0^\circ < \theta < 360^\circ$.

Question 30.

a) Solve the equation

$$3\cos(x - 45^\circ) = 2\sin(x + 60^\circ)$$

for $0^\circ < x < 360^\circ$.

b) Hence solve the equation

$$3\cos(3\theta - 45^\circ) = 2\sin(3\theta + 60^\circ)$$

for $0^\circ < \theta < 120^\circ$.

Question 31. It is given that $\sin x = \frac{2}{\sqrt{14}}$ and $\cos y = \frac{3}{7}$ and that x is obtuse and y is acute. Show that the exact value of $\tan(x - y)$ is $\frac{13\sqrt{10}}{5}$.

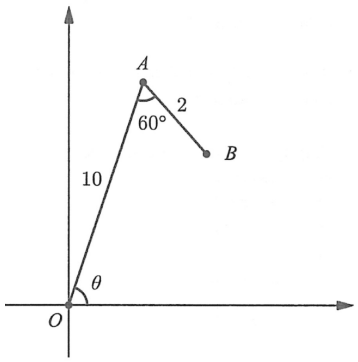
Question 32. It is given that $\tan x = \frac{1}{2}$ and that x is an acute angle. Given that $\tan(x + y) = 13$, find the exact value of $\tan y$.

Question 33. Prove the following trigonometric identity

$$2\sin(x+45°)\sin(x-45°) \equiv \sin^2 x - \cos^2 x.$$

Question 34. The diagram below shows the rod OAB which is pivoted at O. The length of OA is 10 units and the length of AB is 2 units. Taking O as the point with coordinates $(0,0)$, show that the coordinates of B are

$$(9\cos\theta + \sqrt{3}\sin\theta,\ 9\sin\theta - \sqrt{3}\cos\theta).$$



Question 35. Prerequisite knowledge for Q35–Q36: Small Angle Approximation By considering compound angle formulae, show that

$$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

Hence, prove by first principles that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) = -\sin x.$$

Question 36. By considering compound angle formulae, show that

$$\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

Hence, prove by first principles that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x.$$